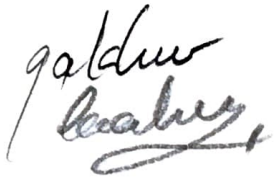
On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

Gökhan Koçmarlı 150720821



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$$P[1\text{ head}] = \binom{4}{1}\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^3 = \frac{4!}{3!1!}\left(\frac{1}{2}\right)^4 = 4\cdot\frac{1}{16} = \frac{1}{4}$$

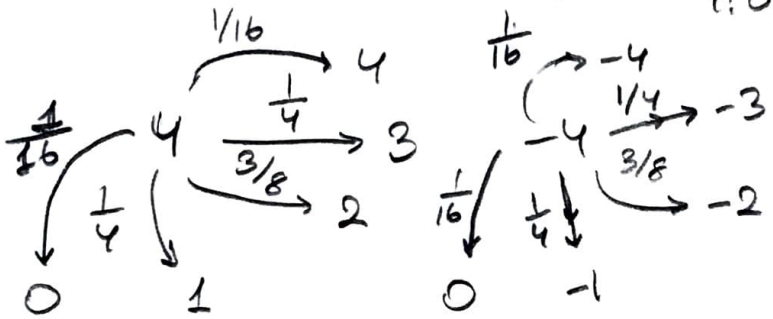
$$P[0\text{ head}] = \binom{4}{0}\left(\frac{1}{2}\right)^0\left(\frac{1}{2}\right)^4 = \frac{4!}{4!0!}\left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

$$P[2\text{ head}] = \binom{4}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^2 = \frac{4!}{2!2!}\left(\frac{1}{2}\right)^4 = \frac{4\cdot 3}{2}\cdot\frac{1}{16} = \frac{3}{8}$$

$$P[3\text{ head}] = \binom{4}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^1 = \frac{4!}{3!1!}\left(\frac{1}{2}\right)^4 = 4\cdot\frac{1}{16} = \frac{1}{4}$$

$$P[4\text{ head}] = \binom{4}{4}\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^0 = \frac{4!}{4!0!}\left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

Question 1



Sample space is

$$\{4, 3, 2, 1, 0, -1, -2, -3, -4\}$$

Outcomes are for +4 → $\{4, 3, 2, 1, 0\}$

$P[A|B]$ if A: being $\{4,3,2,1,0\}$ B: 4 is transmitted

$$P[4|B] = \frac{P[4 \cap B]}{P[B]} = \frac{\frac{1}{2}\cdot\frac{1}{16}}{\frac{1}{2}} = 1/16$$

Same procedure →

$$P[3|B] = \frac{1}{4} \quad P[2|B] = \frac{3}{8}$$

$$P[1|B] = \frac{1}{4} \quad P[0|B] = \frac{1}{16}$$

Event Y=0, there are two situation in order to recieve a 0 signal. The first one, 4 is transmitted but with probability $\frac{1}{16}$ there is a noise wich lowers its magnetide -4. The second one, -4 is transmitted but with probability 1/16 there is a noise which lowers its magnetide by +4. Being transmitted 4 or -4 is equally likely for receiver. So we can say that its poobably is 0.5. Then, if we recieve no signal we can not know the transmitted signal if it is 4 or -4. For other signals like -2, we can predict that it is -4 signal.

Gökhan Koçmarlı

Question 2

Geometric Probability Law $\rightarrow P[\ell] = (1-p)^{\ell-1} \cdot p$, $p = 1/2$

each half-hour costs 1₺.

So

$$P[\text{time cars parked}] = \left(\frac{1}{2}\right)^{k-1} \cdot \left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^{k}$$, $k :=$ the time we want to check for. (30 min period)

$1₺ \cdot P[\text{time parked}] = \text{cost} \cdot \text{time}$

$= \text{total cost}$

a → The prob. cars will pay k₺ → $1₺ \cdot P[\ell] = 1\left(\frac{1}{2}\right)^{k} = \left(\frac{1}{2}\right)^{k}$

b → if maximum charge 6₺:

6₺ → 6 half hour

$$P[k<6] = \sum_{k=1}^{6} \left(\frac{1}{2}\right)^{k} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} + \frac{1}{2^6}$$

new → $p[k]_{(k<6)} = \dfrac{P[k]}{P[k<6]} = \dfrac{\left(\frac{1}{2}\right)^{k}}{\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} + \frac{1}{2^6}}$

Göktan Kocman

Question 3

a

$p = 10^{-4}$ → if it is failing with $p = 10^{-4}$, it won't fail with

$$(1-p) = 1 - 10^{-4} = 1 - 0,0001$$
$$= 0,9999 \rightarrow \text{for only one disk}$$

Since we have 1000 disks

$(0,9999)^{1000}$ → it is power because the failures are independent.

$$P[A \cap B] = P[A] \cdot P[B]$$

b

$P[A] =$ 1 failure $P[B] =$ 0 failure

	1st day 0	2nd day 1
1 failure	1	0
0 failure	0	0

$$P[x] = \underbrace{(10^{-4})^{1}(0,9999)^{1}}_{\text{one failure one not failure}} \cdot 2 + \underbrace{(0,9999)^{2}}_{\text{no failure in two days}}$$

2 → both days

$$P[x] = 2 \cdot 9999 \cdot 10^{-8} + (0,9999)^{2}$$

Göknur Koçmarlı

Question 4

$X \to \{-4, -3, -2, -1, 0, 1, 2, 3\}$

a

$$\text{mean}(x) = \sum_i x_i P_x[x_i] = \frac{1}{8}\sum_i x_i = \frac{1}{8}(-4) = -0,5 \quad (\text{since } P_x[x_i] = \frac{1}{8} \text{ for all } x_i)$$

$$\text{var}(x) = \sum_i (x_i - E(x))^2 P[x_i] = P[x_i] \sum_i [x_i^2 + E(x)^2 - 2x_i E(x)]$$

($E(x)$: mean; $P[x_i]$: still same)

$$\text{var}(x) = \frac{1}{8}\left[\sum_i x_i^2 + \sum_i E^2(x) - 2E(x)\sum_i x_i\right]$$

$$= \frac{1}{8}\left[(16+9+9+1+1+4+9) + (-0,5)^2 \cdot 8 + 1 \cdot -4\right]$$

$$= \frac{1}{8}[44 + 0,25 \cdot 8 - 4] = \frac{1}{8}[40 + 2] = \frac{42}{8}$$

---

b $y = -x^2 + 6 \to y \to \{-10, -3, 2, 5, 6\}$

x	-4	-3	-2	0	-1	1	2	3
y	-10	-3	2	6	5	5	2	-3

$P[-10] = \frac{1}{8}$

$P[-3] = \frac{2}{8} = \frac{1}{4}$

$P[2] = \frac{2}{8} = \frac{1}{4}$

$P[5] = \frac{2}{8} = \frac{1}{4}$

$P[6] = \frac{1}{8}$

$$\text{mean}(y) = \sum_i x \cdot P_x[x_i]$$

$$= (-10)\cdot\frac{1}{8} + (-3)\cdot\frac{1}{4} + (2)\cdot\frac{1}{4} + (5)\frac{1}{4} + (6)\cdot\frac{1}{8}$$

$$= -\frac{5}{4} - \frac{3}{4} + \frac{2}{4} + \frac{5}{4} + \frac{3}{4} = \frac{2}{4} = \frac{1}{2} \to \text{mean}(y) = 0,5$$

$$\text{var}(y) = \sum_i (x_i - 0,5)^2 P_x[x_i] = (-10,5)^2 \cdot \frac{1}{8} + (-3,5)^2 \cdot \frac{1}{4} + (1,5)^2 \cdot \frac{1}{4} + (4,5)^2 \cdot \frac{1}{4} + (5,5)^2 \cdot \frac{1}{8}$$

$$= \frac{1}{8}\left[(-10,5)^2 + 2\cdot(-3,5)^2 + 2\cdot(1,5)^2 + 2(4,5)^2 + (5,5)^2\right]$$

Göktan Kocaman